EDA Report - Student Performance

Abstract

This report explores patterns in a student performance dataset of 120 records using EDA techniques to surface factors that drive final exam scores.

Introduction

Exploratory Data Analysis (EDA) is the practice of summarizing datasets visually and statistically before modeling. We apply it here to academic performance.

Dataset Description

8 columns: Student_ID, Gender, Study_Hours, Attendance, Previous_Scores, Sleep_Hours, Internet_Usage, Final_Score (target). 120 rows, no missing values.

Data Cleaning

Checked types, missing values, duplicates. The dataset was clean; numeric ranges validated within expected bounds.

Exploratory Data Analysis

Computed mean, median, std, min, max for each numeric feature. Distributions are roughly normal for Final_Score.

Visualizations

Histogram of Final_Score, bar chart by Gender, scatter of Study_Hours vs Final_Score, box plots for outlier detection, and a correlation heatmap.

Correlation Analysis

Study_Hours (r~0.65) and Attendance (r~0.45) are the strongest positive correlates with Final_Score. Internet_Usage shows mild negative correlation.

Findings

Students with higher study hours and attendance consistently score higher. Previous scores remain a strong baseline predictor.

Conclusion

Encouraging consistent study habits, regular attendance, and limiting recreational internet use can meaningfully improve student outcomes.